

DoLR · LARR Act 2013 Predictive Analytics System

Smart India Hackathon (SIH26017) · Ministry of Rural Development — Department of Land Resources

System Scope: An AI-powered decision-support and early-detection platform engineered under the *Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (LARR Act 2013)*. It operationalizes the 4-step governance paradigm: **Predict** → **Explain** → **Prioritize** → **Act**.

1. Detailed UI Architecture & Visual Capabilities

The web application features an executive-grade dashboard structured into five dedicated operational consoles:

UI Console / Module	What It Explains & Visualizes	Statutory & Operational Purpose
1. Executive Top KPI Metrics Bar	- Total monitored footprint (Acreage in ha & active parcels) - Capital outlay under monitoring (■ Cr) - Mean expected delay in days (Survival Head output) - Active Section 19(2) statutory lapsing danger counter - Portfolio risk distribution gauge (Low, Medium, Critical)	Gives leadership an instant macro view of systemic delay exposure and capital at risk across national portfolios.
2. Geospatial GIS Map (PostGIS Layer)	- Interactive geographical projection across India's states & districts - Real-time coordinate pins color-coded by delay risk - Radar animation on Critical hazard projects - Stage-wise filtering and inspection drawer	Enables spatial clustering and geographical root-cause analysis (e.g., identifying state/district level bottleneck zones).
3. Section 19(2) 12-Month Lapsing Tracker	365-day statutory countdown clocks per project - Days elapsed since Section 11 preliminary notice - Critical red alerts for projects exceeding 300 days without Section 19 declaration	Enforces strict compliance with Section 19(2) to prevent notifications from automatically lapsing and invalidating acquisitions.
4. TreeSHAP Attribution Diagnostics	- Exact Shapley feature attributions computed directly from tree paths - Categorization into 5 driver families (Compensation, Litigation, Administrative, R&R,, Documentation) - Waterfall visualization of risk-increasing vs risk-mitigating factors	Replaces 'black box' machine learning with transparent, court-defensible explanations for administrative decisions.
5. Counterfactual What-If Sandbox	- Interactive toggle levers for actionable remedies (Sec 26 Direct Negotiations, Surveyor Deputation, Sec 15 Hearing Officers) - Real-time forecast of risk reduction % and delay days saved	Empowers administrators to test the impact of interventions before committing capital and field resources.
6. Dynamic Legal RAG Knowledge Base	384-dimensional vector semantic search across court stays & objections - Drag-and-drop document uploader with automatic chunking & vectorization	Provides instant legal precedent grounding and enables the knowledge base to dynamically absorb court orders in real time.

2. Operational Workflows & User Persona Guide

A. District Collector & Land Acquisition Officer (LAO):

- Monitor the **Sec 19(2) Lapsing Tracker** daily to identify parcels nearing the 365-day legal expiration.
- When a project shows high risk due to Circle Rate Disparity, navigate to **Prescriptive Remedies** to invoke Section 26 Proviso 4 direct negotiation procedures.
- If staff shortages are detected, requisition deputed survey teams to clear Section 12 boundary demarcation.

B. State Ministry & DoLR Policy Monitoring Cell:

1. Utilize the **Geospatial GIS Map** to filter projects by Sector and identify regional litigation trends.
2. Use the **What-If Simulator** to determine which state-level administrative policy shifts (e.g. standardizing Lok Adalat conciliation) yield the greatest overall delay reductions.

C. Project Directors (NHAI, Railways, Energy):

1. Query the **Legal RAG Knowledge Base** for judicial stay precedents regarding multi-crop irrigated land exemptions (Section 10).
2. Upload newly filed citizen objections or environmental clearance letters to continuously update the project's vector risk profile.

3. Technical Justification: Why XGBoost Classifier?

- 1. Heterogeneous Tabular Data Mastery:** Land acquisition records combine continuous financial metrics (cost, circle rate disparity), integer parcel counts, categorical stages, and ratio distributions. Gradient-boosted decision trees (XGBoost) consistently outperform deep neural networks on tabular datasets without requiring artificial normalization.
- 2. Sharp Statutory Threshold Modeling:** Statutory rules under the LARR Act 2013 involve non-linear step functions (e.g., Section 19 lapsing past 365 days, Section 10 multi-crop limits, 70%/80% consent hurdles). Tree-based recursive partitioning captures these hard legal thresholds with mathematical precision.
- 3. Exact & Fast TreeSHAP Integration:** XGBoost allows exact polynomial-time computation of TreeSHAP values (via `pred_contribs=True`), enabling instantaneous feature attribution at serving speed without approximation variance.
- 4. Class Imbalance Resilience:** Critical delay projects represent a specialized distribution. XGBoost's `scale_pos_weight` and focal loss parameters allow the model to heavily prioritize recall on High/Critical risk projects, ensuring zero silent catastrophic lapses.

4. Technical Justification: Why Supabase, pgvector & PostGIS?

- 1. pgvector (384-Dimension HNSW Indexing):** Instead of managing disconnected vector databases (e.g. Pinecone/Chroma), Supabase provides in-database vector storage. By pairing 384D SentenceTransformer embeddings with Hierarchical Navigable Small World (HNSW) cosine indexing, legal similarity queries execute within milliseconds directly inside PostgreSQL.
- 2. PostGIS Native Spatial Geometry:** Land acquisition is inherently geospatial. PostGIS allows storing polygon cadastre maps and point coordinates, enabling spatial boundary queries, regional choropleths, and distance-to-infrastructure calculations directly in SQL.
- 3. Relational Integrity & ACID Milestones:** LARR Act compliance requires strict chronological foreign key relationships between Projects, Section 15 Hearing objections, Compensation disbursements, and Audit logs.
- 4. Scalable Transaction Pooling (Port 6543):** Supabase's built-in PgBouncer transaction pooler prevents connection exhaustion during concurrent batch portfolio scoring and API traffic spikes.